# EXPERIMENT 9

## Python Code Generation using Structured Prompting

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| Name | K Lokesh |
| Register Number | 192472020 |

## AIM

To demonstrate structured prompting for a programming problem and validate the syntax of the generated Python code.

## PROCEDURE

1. Import the ast module for Python syntax validation.
2. Define a prompt describing the programming problem.
3. Prepare a sample Python program to add two numbers.
4. Display the prompt and generated code.
5. Use ast.parse() to validate the Python syntax.
6. Display whether the code validation is successful.

## PROGRAM

```
import ast

print("Python Code Generation using Structured Prompting")

prompt = "Write a Python program to add two numbers."

generated_code = """a = 10
b = 20
total = a + b
print("Sum =", total)
"""

print("\nStructured Prompt:")
print(prompt)

print("\nGenerated Python Code:")
print(generated_code)

try:
    ast.parse(generated_code)
    print("Code Validation: Successful")
except SyntaxError:
    print("Code Validation: Failed")
```

**OUTPUT**

```
print("\nPrompt:", prompt)

code = """
a = 10
b = 20
sum = a + b
print("Sum =", sum)
"""

print("\nGenerated Python Code:")
print(code)

print("Code Validation: Successful")
```

```
Python Code Generation using Structured Prompting

Prompt: Write a program to add two numbers

Generated Python Code:

a = 10
b = 20
sum = a + b
print("Sum =", sum)

Code Validation: Successful
```

**RESULT**

Thus, the structured prompting demonstration and Python code syntax validation were completed successfully.